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Improving hydrolysis acidification by limited aeration in the pretreatment of petrochemical wastewater

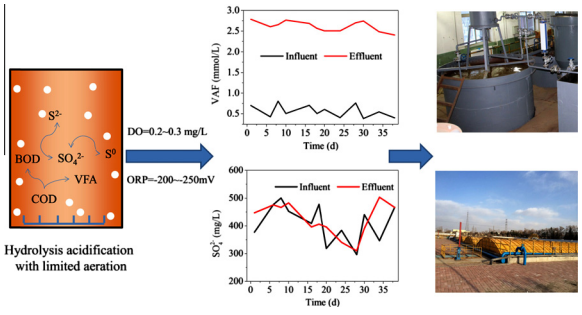
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HIGHLIGHTS

- The optimized DO of limited aeration was 0.2–0.3 mg/L.
- The biodegradability, toxicity and treatability improved obviously.
- SO_4^{2-} reduction can be inhibited by limited aeration hydrolysis acidification.
- SRB was inhibited by limited aeration hydrolysis acidification.

GRAPHICAL ABSTRACT



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ABSTRACT

Petrochemical wastewater was pretreated by hydrolysis acidification to improve the biodegradation and treatability on limited aeration conditions. The results showed limited aeration with DO from 0.2 to 0.3 mg/L (average ORP was −210 mV) was the best condition. The BOD_5/COD of influent was 0.23, and it increased to 0.43 on this condition. Limited aeration can obviously reduce the reduction of SO_4^{2-}, reducing the generation of toxic gas H_2S, and almost no H_2S can be detected in the off-gas. The sulfate reducing bacteria (SRB) diversity and abundance on limited aeration condition was obviously inhibited. Limited aeration condition was benefit for the removal of benzene ring organics, such as benzene, toluene, ethylbenzene and xylenes (BTEX), improving the toxicity and treatability of the wastewater. Based on the experiment results, an anaerobic hydrolysis acidification tank (100,000 m^3) has been transformed into limited aeration hydrolysis acidification tank and it runs well.

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1. Introduction

The petrochemical wastewater is characteristics with high pollutants concentration and salinity, a certain degree of toxicity, low biodegradability and large fluctuations in water quality and quantity. In China, the current annual discharge of industrial wastewater is more than 2.1×10^{10} t, and the percentage of discharged petrochemical wastewater is about 3–4%. However, the percentage of the volatile phenols discharged by petrochemical wastewater is over 35% (Wu et al., 2015). Many technologies, such as biological processes, advanced oxidation, membrane and adsorption methods, can be used in petrochemical wastewater treatment (Lei et al., 2010). Normally, the cost of physical and chemical technologies is relatively high; therefore, the biological wastewater treatment technology is the most promising treatment

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process for petrochemical wastewater treatment because of its affordable cost and high efficiency.

Pretreatment of petrochemical wastewater is often needed due to its low biodegradability and high toxicity. Pretreatment can improve the biodegradability of the wastewater (typically measured with BOD_5/COD), making it suitable for the subsequent biological treatment. In practice, hydrolysis acidification process is often used as a pretreatment process in refractory biodegradable wastewater treatment (Lei et al., 2010). The role of hydrolysis acidification is to change the refractory complex macromolecules, such as aromatic hydrocarbons or heterocyclic substances, into small molecules, such as small molecule organic acids and alcohol. However, the composition of petrochemical wastewater is very complex. The hydrolysis process conditions influence the molecular weight and structure of organics in the effluent, which in turn affects the toxicity of the wastewater (Shi et al., 2011). This may affect the performance of subsequent aerobic biological treatment process. It is known that the hydrolysis of organic-complex wastewater is a limiting stage of the degradation process. In addition to the COD removal and BOD variation, the variation of molecular weight, removal of toxic organic pollutants, reduction of toxicity and biodegradability of wastewater, are also needed to investigate.

The content of sulfate is relatively high in the petrochemical wastewater (Zhang et al., 2013a). The sulfate can be reduced on the hydrolysis acidification condition due to the presence of SRB (Zhang et al., 2013a). Some relatively easily degraded carbon, such as volatile fatty acids (VFA) can be consumed during the process, weakening the role of hydrolysis acidification (Xu et al., 2014). In addition, sulfide from the sulfate reduction also causes toxic effect on microbial metabolism and the emission of hydrogen sulfide is dangerous to the wastewater treatment plant (WWTP) operatives (Zhang et al., 2011; Xu et al., 2012). Therefore, the inhibition of sulfate reduction is very important in the hydrolysis acidification process in the pretreatment of petrochemical wastewater.

There are many ways to inhibit the reduction of sulfate on the reducing conditions, such as ferric iron dosing (Zhang et al., 2009) or nitrate dosing (Chen et al., 2009). Aeration can also inhibit the growth of SRB (Xu et al., 2012). Xu et al. (2012) reported that the activities of SRB were inhibited when the DO was over 0.3 mg/L in a micro-aerobic sulfur recovery reactor. The sulfate reduction decreased obviously in this DO level. The DO level is a promising controlling factor for the sulfate reduction and low DO benefit the fermentative strains. However, the excessive oxygen supply can also affect the hydrolysis acidification process, and it can also stimulate the reduction of biodegradable COD by heterotrophic bacteria oxidation process. The control range of DO is the key to the hydrolysis acidification of petrochemical wastewater. The relationship of SRB and sulfate reduction, and the microbial populations and functional genes distribution in a sulfate removal bioreactor, have been investigated in detail (Xu et al., 2014). However, the aim of limited aeration in their study is to improve the S^0 production through the coupling of sulfate reduction and sulfide oxidation. When the wastewater is the toxic petrochemical wastewater, the effect of limited aeration on hydrolysis acidification performance, concerning the VFA production, sulfate reduction, organic molecular weight variation, toxic organic pollutants removal and toxicity variation, were not clear. The bioreactor performance and the microbial population variation are closely related (Xu et al., 2014). It is necessary to investigate the microbial population variation on different conditions in limited aeration hydrolysis acidification process.

In this study, a continuous operated reactor was run under different DOs. As a comparison, another reactor with the same size was operated on anaerobic condition. The COD removal, VFA production and SO_4^{2-} variation were investigated. In addition, the microbial community structure, organics removal, toxicity and biodegradability variations were also studied.

2. Methods

2.1. Reactor and operation

In this study, two reactors were operated in parallel. The size of two reactors is the same, with the inner diameter of 200 mm and the height of 450 mm. The schematic diagram is shown in Fig. 1. One reactor was operated on limited aeration condition (Reactor A) with different DO, and the wastewater and activated sludge were mixed by aeration and stirred mixer. The other was operated on anaerobic condition (Reactor B), with wastewater and activated sludge mixed by slowly operated stirred mixer. Each reactor was equipped with a settling tank used for wastewater and activated sludge separation, as well as sludge recycle. The influent of the Reactor A and B was the same with the flowrate of 0.875 L/h, resulting in the hydraulic retention time (HRT) of 16 h. The average mixed liquid suspended solids (MLSS) in Reactor A and B was 6–8 g/L. Both reactors were equipped with DO, ORP and pH meters. The whole operating period was about 135 d. There was no change in the operating parameters in Reactor B. The average ORP of Reactor B was −350 mV. However, in the operation of Reactor A, the whole time was divided into 4 periods (I, II, III and IV), with the average DO of about 0.1–0.2 (I), 0.2–0.3 (II), 0.3–0.4 (III) and 0.4–0.5 mg/L (IV), respectively (Supplementary data, Fig. S1). The average ORP of Period I, Period II, Period III and Period IV were −270, −210, −164 and −99 mV, respectively (Supplementary data, Fig. S2).

2.2. Wastewater

The wastewater used in this study was the influent of a centralized petrochemical WWTP with hydrolysis acidification-anoxic/oxic as the core process. The wastewater treated by this petrochemical WWTP is the mixed wastewater discharged from a petrochemical industrial park containing more than 50 sets of petrochemical production plants, including typical petroleum refining plants, basic petrochemical raw materials (intermediates) production plants and petrochemical synthetic material plants. The main qualities of the wastewater were: COD 320–500 mg/L, BOD_5 50–170 mg/L, NH_4^+—N 15–25 mg/L, NO_3^-—N 2.5–12.1 mg/L, PO_4^{3-}—P 2.0–7.0 mg/L, SO_4^{2-} 296–570 mg/L, and pH 6.5–8.7.

2.3. Water quality analysis

COD, BOD_5 were measured according to the Standard Methods (APHA, 2005). DO and ORP were determined by a portable meter (WTW 340i, Germany). VFA was measured by titration method using Delta 320-s pH meter (Mettler-Toledo Group, Greifensee, Switzerland) (Feitkenhauer et al., 2002). Sulfate was determined by ion chromatography using a conductivity detector (Dionex ICS-1000, Japan). Total sulfide in the wastewater was analyzed by the potentiometric titration method (Mettler-Toledo Group, Greifensee, Switzerland). H_2S in the off-gas was measured by gas chromatography (Agilent 6890, USA) according to the Standard Methods (APHA, 2005).

2.4. Microbial community structure analysis

The activated sludge samples were taken on day 50 from Reactor A (DO = 0.2–0.3 mg/L) and B. Total DNA extraction

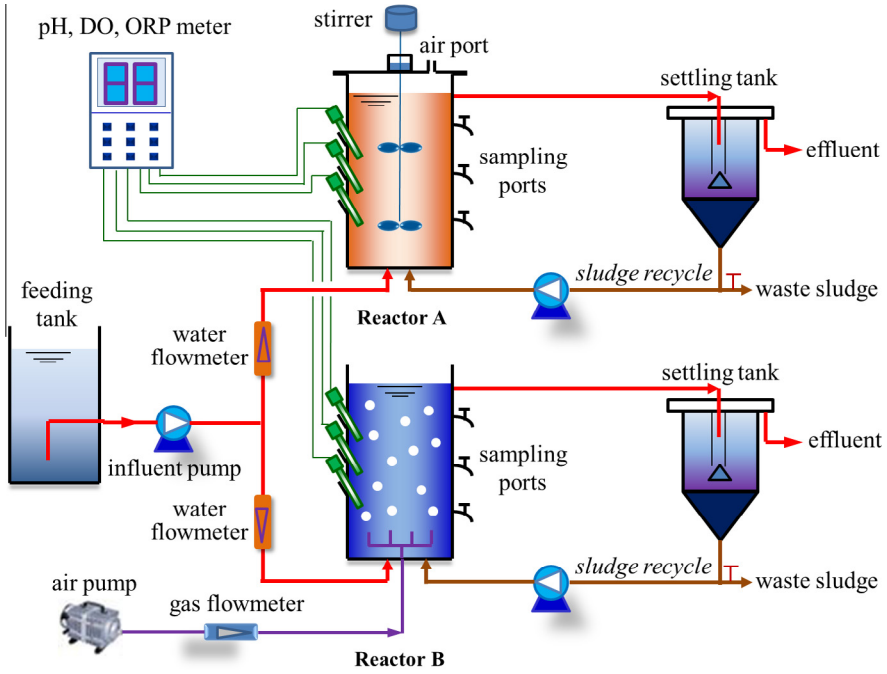


Fig. 1. The schematic diagram of the reactors.

(100 μL) from the digestate sludge samples was carried out using Mo-Bio PowerSoil DNA Isolation Kit (MoBio Laboratories, Inc., USA) according to the manufacturer's instructions. The quality of the extracted DNA was evaluated by electrophoresis on a 1% agarose gel. To determine the added amount of DNA for PCR reaction, the Qubit 2.0 DNA detection kit was used for precise quantification of genomic DNA. The universal primer for sequencing has been added into the primers used for PCR amplification. The primers and amplification conditions were conducted according to the description of Wan et al. (2014). A 454 Life Sciences Genome Sequencer FLX Titanium instrument (Roche) was used for sequencing in this study. Mothur (Version 1.30.1) was used for the sequencing data analysis.

2.5. OUR and biodegradability test

Oxygen uptake rate (OUR) was measured to assess the toxicity of influent, limited aeration and anaerobic effluents. The wastewater samples were collected on day 50. The measurement was carried out according to ISO 8192 (2007). Specific OUR (SOUR) was used to evaluate the toxicity of different wastewaters.

The biodegradability of wastewater was measured by aeration batch experiment. 1 L influent, limited aeration effluent and anaerobic effluent were taken and then filled into 3 aeration reactors (2 L each), respectively. Activated sludge was taken from the aeration tank of the petrochemical WWTP which the wastewater was used in this study and then added them into the 3 reactors with the MLSS concentration of about 4 g/L. Adequate oxygen were supplied to the 3 reactors with the same air flowrate of 0.5 m^3/h. The total aeration time was 72 h. The samples were taken from the 3 reactors and COD was measured.

2.6. Measurement of organic pollutants

Volatile organic compounds (VOCs) and semi-volatile organic compounds (SVOCs) were measured in this study (influent, limited aeration effluent and anaerobic effluent samples were collected on day 50). The VOCs were measured according to the EPA 524.2 using purge and trap coupled with gas chromatography/mass spectrometry (GC/MS) (EPA 524.2, 1995). SVOCs were measured using dichloromethane liquid–liquid extraction coupled with GC/MS. The preparation of the samples was the same as our previous study (Lai et al., 2012). The identification of the organics was based on the NIST 05 mass spectral library database. According to the match results, 48 organic standards were used to furtherly identify the substance by the retention time of mass spectra comparison. Finally, GC-FID was used for the quantitative determination of BTEX and phenol.

3. Results and discussion

3.1. The performance of limited aeration and anaerobic reactors

3.1.1. COD removal and variation of BOD_5/COD

The average COD of the influent was 421 mg/L during the experiment. The average COD of the Reactor B effluent was 375 mg/L and the removal rate was about 10.9%. Wang et al. (2014) reported that the COD removal rates of hydrolysis acidification were about 26.9%, 28% and 30–40% when treating sweet potato starch wastewater, Chinese traditional medicine wastewater and tannery wastewater, respectively. The removal rate in this study is lower than some traditional hydrolysis acidification process. It may be due to the toxicity of the influent as it contained many toxic organic pollutants. This can be confirmed in the following section that presents the toxicity and biodegradability. The COD removal in Reactor A was higher than that in Reactor B. The average COD of the Reactor A effluent was 314 mg/L with the removal rate of 25.4%. The COD removal rates, which increased with the increase of DO, were 16.2%, 24.9%, 27.9% and 31.6% in Period I, Period II, Period III and Period IV, respectively (Supplementary data, Fig. S3).

The purpose of the hydrolysis acidification is not to removal maximum COD, but to enhance the BOD level or BOD_5/COD value (Wang et al., 2014). The BOD_5/COD value is an important index for determining the biological treatment process. Generally, biological treatment process is not appropriate when the BOD_5/COD value is lower than 0.3. The BOD_5/COD value of influent was only

0.23, which was even lower than the tannery wastewater (Wang et al., 2014). The value increased to 0.35 when treated by Reactor B. In the limited aeration reactor, the BOD_5/COD value was 0.41 in Period I, indicating that biodegradability can be enhanced by limited aeration. The value increased continuously to 0.43 in Period II with the increase of aeration. However, the value decreased quickly to 0.31 and 0.20 in Period III and Period IV When continued to increase the amount of aeration, as shown in Fig. 2.

3.1.2. The production of VFA

The VFA of the influent was approximately 0.62 mmol/L. Some VFA in the influent may be produced during the long distance transportation of wastewater inside the pipeline, and some petrochemical wastewater contains low concentration of VFA, such as acetaldehyde production wastewater. The production of VFA in Reactor B was relatively stable during the experiment, as shown in Fig. 3. The average VFA concentration in Reactor B effluent was 2.37 mmol/L. More than 1.75 mmol/L VFA was produced during anaerobic hydrolysis acidification process. VFA are easily degradable carbon sources, therefore, hydrolysis acidification process can enhance the biodegradability of the wastewater, and this is consistent with the change of BOD_5/COD, as shown in Fig 2. The production of VFA in Reactor A was different from Reactor B. The concentration of VFA in Period I was 2.41 mmol/L, which is higher than that in Reactor B. With the increase of aeration, the VFA in Period II was 2.57 mmol/L, which was higher than that in Period I, indicating that limited aeration (DO = 0.2–0.3 mg/L) can improve the hydrolysis acidification compared to anaerobic hydrolysis acidification. However, the production of VFA decreased significantly when DO was higher than 0.3 mg/L. The VFA in Period III and Period IV were 1.68 and 1.01 mg/L, which were lower than that in Reactor B. This is because the VFA can be oxidized by the excess oxygen when applying more air. The production of VFA in this study was lower than some results reported previously (Rajagopal and Béline, 2011). It is reported that the increase of VFA/bicarbonate alkalinity ratio can enhance the VFA accumulation. However, the influent was relatively stable without any changes in pH. The difference may be due to the toxicity of the petrochemical wastewater.

3.1.3. The variation of SO_4^{2-}

Industrial wastewaters usually contain high levels of SO_4^{2-} than municipal wastewater (Zhang et al., 2013a). In this study, the

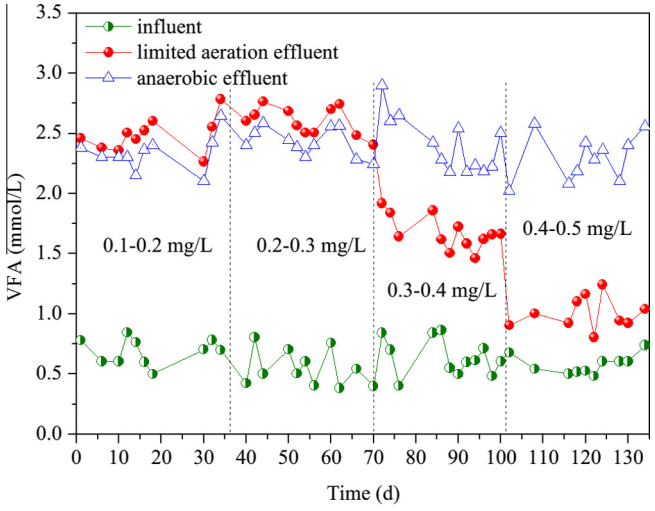


Fig. 3. The production of VFA in limited aeration and anaerobic effluents.

average SO_4^{2-} concentration in the wastewater was 470.8 mg/L. SO_4^{2-} can be reduced into S^{2-} or S by SRB in the anaerobic condition (Chen et al., 2009). Fig. 4 is the variations of SO_4^{2-} in influent, limited aeration and anaerobic effluents during the experiment. As shown in Fig. 4, there was obvious reduction of SO_4^{2-} in anaerobic effluent. The average SO_4^{2-} concentration in the anaerobic effluent was 336.8 mg/L, with the removal rate of 28.5%. The SO_4^{2-} concentrations in the limited aeration effluent were 412.1, 407.0, 525.4 and 567.9 mg/L in Period I, Period II, Period III and Period IV, respectively. The reduction of SO_4^{2-} on limited aeration condition was inhibited compared to anaerobic condition. The removal rate was only 5.6% in Period I and almost no SO_4^{2-} was removed in Period II. However, the SO_4^{2-} concentration in effluent was higher than that in influent in Period III and Period IV. The effluent SO_4^{2-} concentration increased with the increase of DO level. This is consistent with the previous study. Xu et al. (2012) reported that SO_4^{2-} reduction can be inhibited when DO was over 0.3 mg/L. This value is higher than that in this study (0.2–0.3 mg/L) as the wastewater was different. In addition, the oxidation of some sulfur-containing organics would release part SO_4^{2-} into the

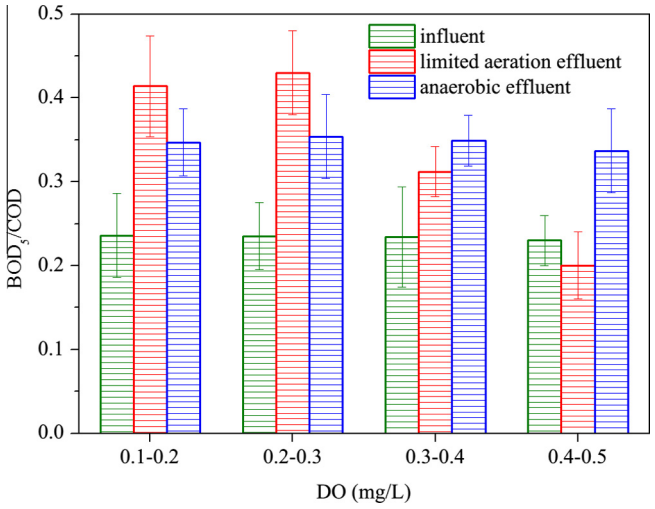


Fig. 2. The variation of BOD_5/COD in influent, limited aeration and anaerobic effluents.

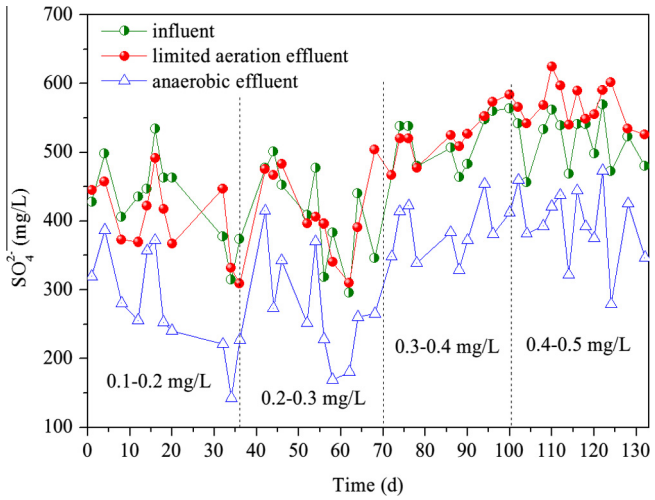


Fig. 4. The variations of SO_4^{2-} in influent, limited aeration and anaerobic effluents.

wastewater (Li et al., 2015), causing the increase of SO_4^{2-} in Period III and Period IV.

3.2. Microbial community structure

The microbial community structure is closely related to the operation condition of the reactor (Yadav et al., 2014). The observed OTUs in reactor A (DO = 0.2–0.3 mg/L) and B samples were 3027 and 2806, respectively. The abundances of different phylum in two reactor's sludge were shown in Fig. S4 (Supplementary data). The result showed that the microbial community structure of the two reactors was similar. *Proteobacteria*, *Chloroflexi*, *Firmicutes*, *Bacteroidetes*, *Planctomycetes*, *Acidobacteria*, *Deferribacteres* and *Actinobacterium* were the dominant microorganisms in phylum level in the two reactors. More than 35% of the OTUs detected in both rectors was *Proteobacteria*. *Proteobacteria* and *Firmicutes* are facultative bacteria, which are the most abundant phylum in most wastewater treatment bioreactors (Liang et al., 2014, 2015). They can degrade organics in the WWTP (Zhang et al., 2012). It is reported that *Firmicutes* can be present in toxic and refractory wastewater treatment system (Liang et al., 2014). Most of *Chloroflexi* are filamentous bacteria, and they have the ability to degrade macromolecular organics (Mielczarek et al., 2012). The *Bacteroidetes* and *Acidobacteria* have been reported that they can widespread in fermentation and acidification bioreactors (Liang et al., 2015). The carbohydrates can be hydrolyzed by these bacteria into monosaccharides, and then monosaccharides can be furtherly fermented into lactic acid, acetic acid, formic acid or pyruvic acid. They can also hydrolyze the protein into amino acids and organic acids, or lipids into small molecule fatty acids and alcohol. This is consistent with the function of hydrolysis acidification of the two reactors. *Actinobacterium* has been reported that they have the ability to degrade the macromolecule and toxic organics (Wang et al., 2010). The abundance of them in limited aeration reactor was higher than that in anaerobic reactor. The distribution of microorganisms was closely related to the performance of the reactors.

Early studies suggest that SRB are strictly obligate anaerobes, but recent findings show that SRB can survive in the presence of low molecular oxygen (Ontiveros-Valencia et al., 2014). This is consistent with this study. The relative abundance of the dominant SRB at genus level in Reactor A and B was shown in Fig. 5. The total number of SRB at genus level detected in Reactor A and B were 10 and 18, respectively. The total relative abundance of SRB at genus level in Reactor A was 0.87%. However, the value was 2.75% in Reactor B. The SRB diversity and abundance in Reactor A was lower than that in Reactor B, indicating that the SRB were obviously inhibited by the limited aeration condition when DO was between 0.2 and 0.3 mg/L. This is consistent with the SO_4^{2-} reduction performance in the two reactors.

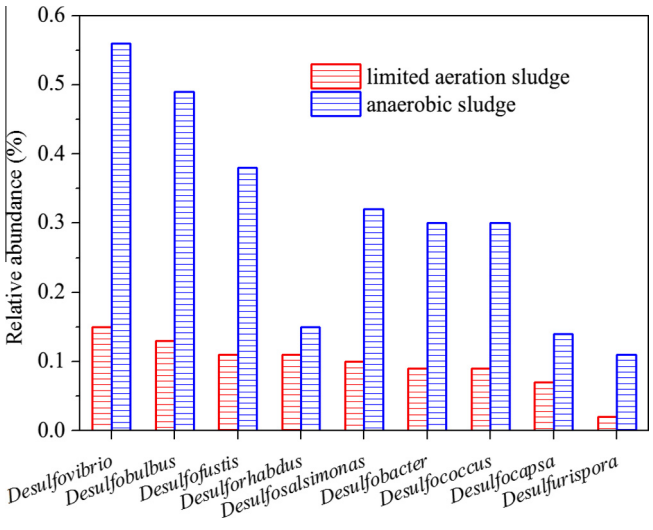


Fig. 5. Relative abundance of the dominant SRB at genus level.

3.3. Enhancement of hydrolysis acidification by limited aeration

The reduction process of SO_4^{2-} needs electron donor, such as readily biodegradable organics (usually VFA and some other small molecule organics), to accomplish (Wang et al., 2008). The utilization of SRB for SO_4^{2-} reduction can contribute parts of biodegradable COD removal (Wang et al., 2008; Zhang et al., 2013a). The total reduction process of SO_4^{2-} is as the following (van Houten et al., 1997):

$$SO_4^{2-} + ATP + 8e^- + 10H^+ \rightarrow H_2S + 5H_2O + 2Pi + AMP \quad (1)$$

Take limited aeration Period II for example, there was almost no biodegradable COD consumed by sulfate reduction on this condition, however, the theoretical biodegradable COD consumed by SO_4^{2-} reduction during the anaerobic condition was about 90.4 mg/L. In other words, in theory, the same amount of biodegradable COD can be saved on the limited aeration condition (DO = 0.2–0.3 mg/L) simultaneously. However, due to the existence of oxygen, part of biodegradable COD can be oxidized. Generally speaking, from the results it showed that the inhibition of SO_4^{2-} reduction can save the utilization of the generated biodegradable COD. Therefore, the weak or block of SO_4^{2-} reduction to some degree by low DO condition in hydrolysis acidification process can save the consumption of BOD in a certain extent. In this study, the proper DO concentration which can improve the hydrolysis acidification was 0.2–0.3 mg/L. The limited aeration can also benefit for the less generation of toxic gas H_2S, which was produced during the SO_4^{2-} reduction. The H_2S concentration in the anaerobic reactor off-gas was 72.3 ± 14.2 mg/m^3 and almost no H_2S can be detected in the limited aeration reactor off-gas when DO was higher than 0.2 mg/L. H_2S is a strong nerve poison gas (Martinez-Cutillas et al., 2015). From this perspective, limited aeration hydrolysis acidification is a safe and environmentally friendly technology for the hydrolysis acidification pretreatment of petrochemical wastewater.

3.4. The toxicity and biodegradability

OUR is one fundamental index to assess the toxicity of wastewater to activated sludge based on oxygen usage. According to the DO slope (Supplementary data Fig. S5), the SOUR of the influent was 2.05 mg O_2/g-MLSS-h, indicating there was toxicity of the wastewater to the activated sludge (Garcia-Ochoa et al., 2010). The SOUR increased to 9.65 mg O_2/g-MLSS-h after limited aeration hydrolysis acidification on DO 0.2–0.3 mg/L. The toxicity of the petrochemical wastewater decreased obviously, which is consistent with the change of BOD_5/COD. Hydrolysis acidification is very important and essential for the biological petrochemical wastewater treatment. As a comparison, the anaerobic effluent SOUR was 9.10 O_2/g-MLSS-h, which was slightly lower than that in limited aeration effluent.

Biodegradability is one of the important characteristics for wastewater (Wu et al., 2014). The aim of the hydrolysis acidification is to improve the biodegradability of the wastewater. The biodegradable COD can be degraded by biomass in the presence of oxygen, as shown in Fig. S6 (Supplementary data). The

degradation process can be divided into 3 phases. The COD dropped quickly in the first 2 h, due to the absorption of the activated sludge (Ben et al., 2014). In the next 10 h, the COD decreased at a slower speed than the first 2 h by the oxidization of activated sludge. In the next 60 h, the COD decreased slightly with minor fluctuation. Finally, the remained COD of influent, limited aeration effluent and anaerobic effluent samples by biodegradability test were about 102, 75 and 85 mg/L, respectively. The limited aeration hydrolysis acidification can enhance the treatability of the petrochemical wastewater and the effluent is more suitable for the following aerobic biological treatment processes.

In this study, 48 kinds of organics were detected in the influent (data not shown, not included the substances with the peak area less than 5% of the largest peak area), during which the variation of benzene, toluene, ethylbenzene, p-xylene, m-xylene and phenol were shown in Fig. 6. BTEX are the major aromatic and toxic components in petrochemical wastewater (Ei-Naas et al., 2014). Previous studies reported that microorganisms preferentially degrade BTEX under aerobic or microaerobic conditions rather than anaerobic conditions. This is because the metabolic pathways for the degradation of BTEX are provided by dioxygenases and monooxygenases enzymatic systems (Khan et al., 2001). The ring is opened and then some low molecular weight intermediates are produced (Zhang et al., 2013b). These compounds can be furtherly transformed into small molecular weight organics with good biodegradability by hydrolysis acidification. In this study, the total concentration of BTEX and phenol in the influent was 44.61 mg/L. The concentration was 27.36 mg/L in the anaerobic effluent with the removal rate of 38.7%. However, the concentration was only 8.00 mg/L in the limited aeration effluent (DO = 0.2–0.3 mg/L) and the removal rate was up to 82.1%. The reduction and transformation of BTEX or phenol can significantly reduce the toxicity and biodegradation (Mozo et al., 2011), enhancing the treatability of the wastewater. This is consistent with the results of SOUR and biodegradability tests.

The results prove that the limited aeration hydrolysis acidification process is suitable for the pretreatment of petrochemical wastewater. Based on the experiment results, a pilot scale experiment (2 m^3/h) was carried out and the same conclusions were obtained. Finally, an anaerobic hydrolysis acidification tank (100,000 m^3) of a petrochemical WWTP has been reformed into limited aeration hydrolysis acidification tank. The tank runs well after reformation (Supplementary data).

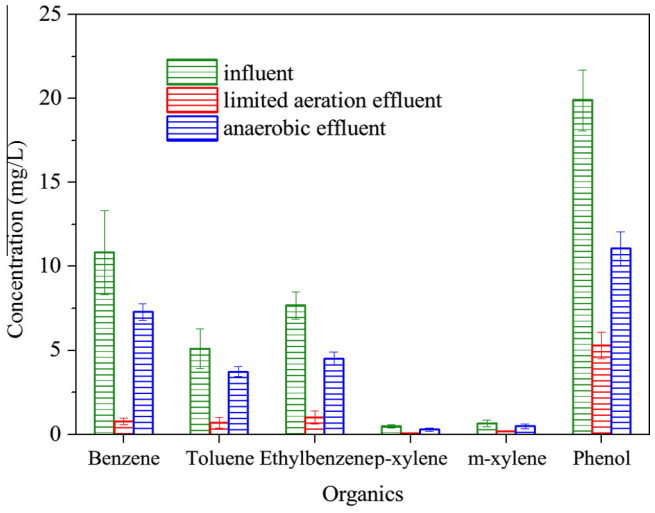


Fig. 6. The typical organics variation in the influent, limited aeration and anaerobic effluents.

4. Conclusions

Limited aeration with DO from 0.2 to 0.3 mg/L (average ORP was −210 mV) is suitable for the hydrolysis acidification of petrochemical wastewater. The COD removal rate was 24.9% when the influent COD was 421 mg/L. The BOD_5/COD can increase from 0.23 to 0.43 on this condition with the VFA concentration of 2.57 mmol/L. No H_2S can be detected in the limited aeration reactor off-gas. Microbial community structure analysis showed that the diversity and abundance of SRB were inhibited obviously on limited aeration condition.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.biortech.2015.06.072.

References

APHA, 2005. Standard Methods for the Examination of Water & Wastewater. American Public Health Association, Washington, DC.
Ben, W., Qiang, Z., Yin, X., Qu, J., Pan, X., 2014. Adsorption behavior of sulfamethazine in an activated sludge process treating swine wastewater. J. Environ. Sci. 26, 1623–1629.
Chen, C., Wang, A., Ren, N., Lee, D.J., Lai, J.Y., 2009. High-rate denitrifying sulfide removal process in expanded granular sludge bed reactor. Bioresour. Technol. 100, 2316–2319.
Ei-Naas, M.H., Acio, J.A., Telib, A.E.E., 2014. Aerobic biodegradation of BTEX: progresses and prospects. J. Environ. Chem. Eng. 2, 1104–1122.
EPA, 1995. Measurement of Purgeable Organic Compounds in Water by Capillary Column Gas Chromatography/Mass Spectrometry. EPA 524.2. U.S. Environmental Protection Agency, Washington, DC.
Feitkenhauer, H., von Sachs, J., Meyer, U., 2002. On-line titration of volatile fatty acids for the process control of anaerobic digestion plants. Water Res. 36, 212–218.
Garcia-Ochoa, F., Gomez, E., Santos, V.E., Merchuk, J.C., 2010. Oxygen uptake rate in microbial processes: an overview. Biochem. Eng. J. 49, 289–307.
ISO 8192, 2007. Water Quality-Test for Inhibition of Oxygen Consumption by Activated Sludge for Carbonaceous and Ammonium Oxidation. International Organization for Standardization, Geneva.
Khan, A.A., Wang, R.F., Cao, W.W., Doerge, D.R., Wennerstrom, D., Cerniglia, C.E., 2001. Molecular cloning, nucleotide sequence, and expression of genes encoding a polycyclic aromatic ring dioxygenase from *Mycobacterium* sp. strain PYR-1. Appl. Environ. Microbiol. 67, 3577–3585.
Lai, B., Zhou, Y., Qin, H., Wu, C., Pang, C., Lian, Y., Xu, J., 2012. Pretreatment of wastewater from acrylonitrile–butadiene–styrene (ABS) resin manufacturing by microelectrolysis. Chem. Eng. J. 179, 1–7.
Lei, G., Ren, H., Ding, L., Wang, F., Zhang, X., 2010. A full-scale biological treatment system application in the treated wastewater of pharmaceutical industrial park. Bioresour. Technol. 101, 5852–5861.
Li, W., Niu, Q., Zhang, H., Tian, Z., Zhang, Y., Gao, Y., Li, Y., Nishimura, O., Yang, M., 2015. UASB treatment of chemical synthesis-based pharmaceutical wastewater containing rich organic sulfur compounds and sulfate and associated microbial characteristics. Chem. Eng. J. 260, 55–63.
Liang, B., Cheng, H., van Nostrand, J.D., Ma, J., Yu, H., Kong, D., Liu, W., Ren, N., Wu, L., Wang, A., Lee, D.J., Zhou, J., 2014. Microbial community structure and function of nitrobenzene reduction biocathode in response to carbon source switchover. Water Res. 54, 137–148.
Liang, S., Gliniewicz, K., Mendes-Soares, H., Settles, M.L., Forney, L.J., Coats, E.R., McDonald, A.G., 2015. Comparative analysis of microbial community of novel lactic acid fermentation inoculated with different undefined mixed cultures. Bioresour. Technol. 179, 268–274.
Martinez-Cutillas, M., Gil, V., Mañé, N., Clavé, P., Gallego, D., Martin, M.T., Jimenez, M., 2015. Potential role of the gaseous mediator hydrogen sulphide (H_2S) in inhibition of human colonic contractility. Pharmacol. Res. 93, 52–63.
Mielczarek, A.T., Kragelund, C., Eriksen, P.S., Nielsen, P.H., 2012. Population dynamics of filamentous bacteria in Danish wastewater treatment plants with nutrient removal. Water Res. 46, 3781–3795.

Mozo, I., Stricot, M., Lesage, N., Spérandio, M., 2011. Fate of hazardous aromatic substances in membrane bioreactors. Water Res. 45, 4551–4561.
Ontiveros-Valencia, A., Tang, Y., Krajmalnik-Brown, R., Rittmann, B.E., 2014. Managing the interactions between sulfate- and perchlorate-reducing bacteria when using hydrogen-fed biofilms to treat a groundwater with a high perchlorate concentration. Water Res. 55, 215–224.
Rajagopal, R., Béline, F., 2011. Anaerobic hydrolysis and acidification of organic substrates: determination of anaerobic hydrolytic potential. Bioresour. Technol. 102, 5653–5658.
Shi, H., Fatehi, P., Xiao, H., Ni, Y., 2011. A combined acidification/PEO flocculation process to improve the lignin removal from the pre-hydrolysis liquor of kraft-based dissolving pulp production process. Bioresour. Technol. 102, 5177–5182.
van Houten, R.T., Yun, S.Y., Lettinga, G., 1997. Thermophilic sulphate and sulphite reduction in lab-scale gas-lift reactors using H_2 and CO_2 as energy and carbon source. Biotechnol. Bioeng. 55, 807–814.
Wan, C.L., Zhang, Q.L., Lee, D.J., Wang, Y.Y., Li, J.N., 2014. Long-term storage of aerobic granules in liquid media: viable but non-culturable status. Bioresour. Technol. 166, 464–470.
Wang, A.J., Ren, N.Q., Wang, X., Lee, D., 2008. Enhanced sulfate reduction with acidogenic sulfate-reducing bacteria. J. Hazard. Mater. 154, 1060–1065.
Wang, Z., Wang, T., Gong, F., Zhang, J., Hong, Q., Li, S., 2010. Biodegradation of carbendazim by a novel actinobacterium *Rhodococcus jialingiae* djl-6-2. Chemosphere 81, 639–644.
Wang, K., Li, W., Gong, X., Li, X., Liu, W., He, C., Wang, Z., Minh, Q.N., Chen, C., Wang, J., 2014. Biological pretreatment of tannery wastewater using a full-scale hydrolysis acidification system. Int. Biodeterior. Biodegrad. 95, 41–45.
Wu, J., Yan, G., Zhou, G., Xu, T., 2014. Wastewater COD biodegradability fractionated by simple physical–chemical analysis. Chem. Eng. J. 258, 450–459.
Wu, C., Gao, Z., Zhou, Y., Liu, M., Song, J., Yu, Y., 2015. Treatment of secondary effluent from a petrochemical wastewater treatment plant by ozonation-biological aerated filter. J. Chem. Technol. Biotechnol. 90, 543–549.
Xu, X., Chen, C., Wang, A., Fang, N., Yuan, Y., Ren, N., Lee, D.J., 2012. Enhanced elementary sulfur recovery in integrated sulfate-reducing, sulfur-producing rector under micro-aerobic condition. Bioresour. Technol. 116, 517–521.
Xu, X., Chen, C., Wang, A., Yu, H., Zhou, X., Guo, H., Yuan, Y., Lee, D.J., Zhou, J., Ren, N., 2014. Bioreactor performance and functional gene analysis of microbial community in a limited-oxygen fed bioreactor for co-reduction of sulfate and nitrate with high organic input. J. Hazard. Mat. 278, 250–257.
Yadav, T.C., Khardenavis, A.A., Kapley, A., 2014. Shifts in microbial community in response to dissolved oxygen levels in activated sludge. Bioresour. Technol. 165, 257–264.
Zhang, L., Keller, J., Yuan, Z., 2009. Inhibition of sulfate-reducing and methanogenic activities of anaerobic sewer biofilms by ferric iron dosing. Water Res. 43, 4123–4132.
Zhang, J.X., Zhang, Y.B., Quan, X., Liu, Y.W., An, X.L., Chen, S., Zhao, H.M., 2011. Bioaugmentation and functional partitioning in a zero valent iron-anaerobic reactor for sulfate-containing wastewater treatment. Chem. Eng. J. 174 (1), 159–165.
Zhang, T., Shao, M.F., Ye, L., 2012. 454 pyrosequencing reveals bacterial diversity of activated sludge from 14 sewage treatment plants. ISME J. 6, 1137–1147.
Zhang, J., Zhang, Y., Chang, J., Quan, X., Li, Q., 2013a. Biological sulfate reduction in the acidogenic phase of anaerobic digestion under dissimilatory Fe(III)-reducing conditions. Water Res. 47, 2033–2040.
Zhang, L., Zhang, C., Cheng, Z., Yao, Y., Chen, J., 2013b. Biodegradation of benzene, toluene, ethylbenzene, and o-xylene by the bacterium *Mycobacterium cosmeticum* byf-4. Chemosphere 90, 1340–1347.